

## **Healthcare Resource Availability: Analyzing Global Disparities**

### **ADAN8880 - Directed Practicum in Analytics**

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## **Dataset Processing Report**

### **Data cleaning & standardization**

To ensure the dataset was accurate and ready for analysis, we applied several data cleaning steps:

- Removed unnecessary columns: 'series code' and 'country code' were dropped since they did not contribute to analysis.
- Standardized column names: converted to lowercase, replaced spaces with underscores, and removed special characters for consistency.
- Identified zero values: we analyzed whether zero values in specific indicators were valid data points or should be treated as missing values.
- Duplicate records were removed to prevent bias in analysis and model building. However, duplicates that represented meaningful distinctions, such as 'healthcare resources' and 'healthcare expenditure' were retained to maintain dataset integrity.
- Certain indicators, such as 'external health expenditure (%)' and 'out-of-pocket health expenditure (%)' contained zeros. We examined whether these zeros were actual recorded values or missing data. Meaningful zeros were retained, and cases where zero values were unlikely were replaced with NaN to indicate missing values.
- Sorted data by country\_name, series\_name, and year to stay in chronological order for time-series modeling.
- Performed a final missing value check to confirm all missing values were addressed before saving the cleaned dataset.

### **Feature engineering**

- To enhance the dataset's usability for analysis and modeling, we created key time-series features:
- Lag features (lag3, lag5) to capture historical patterns by shifting values 3 and 5 years back, which helps to detect trends.

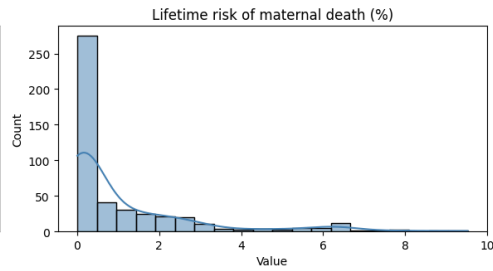
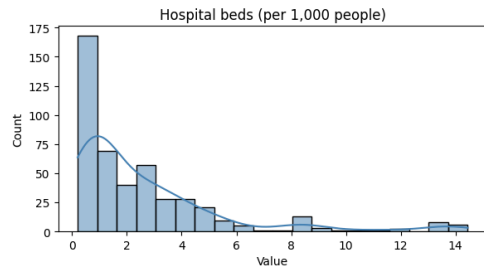
- A 3 year lag helps capture short-term cyclical patterns and policy effects that take a few years to materialize.
- A 5 year lag provides a broader view, and is useful for detecting structural shifts and long-term policy impacts.
- Rolling averages (rolling3, rolling5) to smooth fluctuations and capture long-term trends.
  - A 3 year rolling average to capture recent trends.
  - A 5-year rolling average helps capture longer-term trends, reducing volatility and emphasizing stable patterns over time.
- Year over year growth rate to measure percentage change from the previous year for trends comparable across countries.
- Income group encoding to categorize countries into numeric groups (1 = low, 2 = middle, 3 = high) to make income\_group usable in models.

### **Log transformation for skewed data**

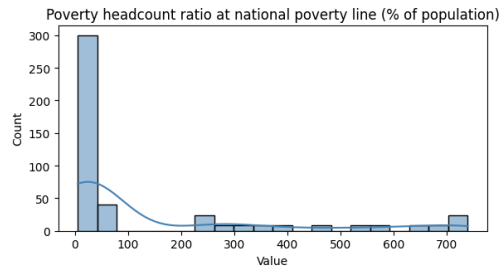
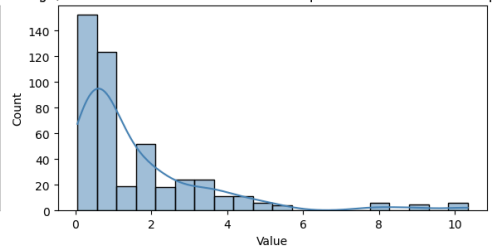
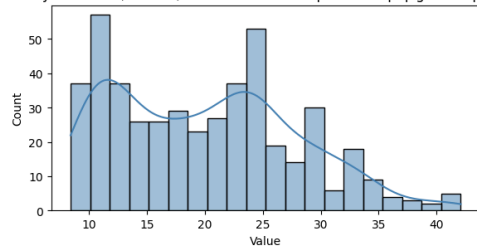
Some indicators exhibited right-skewed distributions, meaning certain countries had extremely high values compared to others. To improve comparability and model performance, we applied a log transformation to:

- 'Hospital beds per 1,000 people' → some countries have very few hospital beds, but others have much higher values, creating a highly skewed distribution. Log transformation helps normalize the spread.
- 'Lifetime risk of maternal death (%)' → many countries have near-zero maternal mortality, but others have significantly higher risks, which leads to an uneven distribution. Log transformation makes the data more comparable.
- 'Mortality from CVD, cancer, diabetes, or CRD (%)' → the distribution is right-skewed with a few countries experiencing very high mortality rates. Log transformation reduces this skewness.
- 'Proportion of population spending >25% of income on healthcare (%)' → many countries have low out-of-pocket expenditures, but some have much higher burdens. Log transformation improves the scale balance.
- 'Poverty headcount ratio at the national poverty line (%)' → this metric varies widely across income groups, which leads to a skewed distribution. Log transformation stabilizes variance and makes patterns more interpretable.

Before applying the transformation, we plotted histograms to examine the skewed distributions

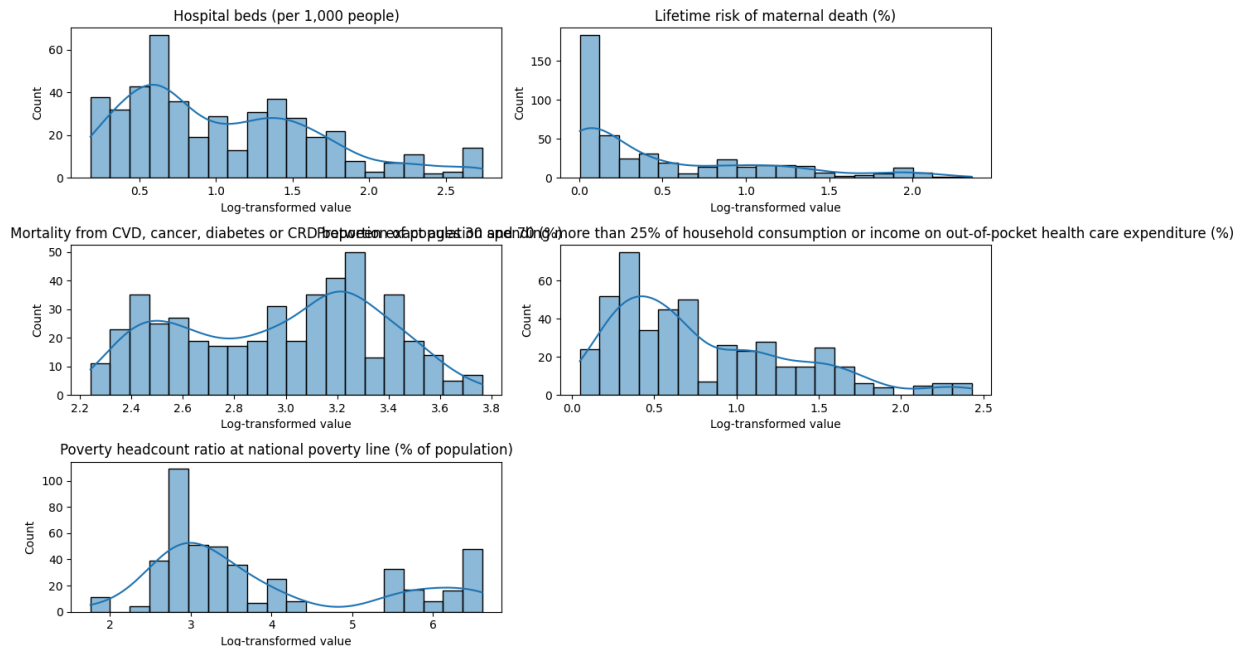


Mortality from CVD, cancer, diabetes or CRD Proportion of population spending more than 25% of household consumption or income on out-of-pocket health care expenditure (%)



- As we have discussed, the above graph before log transformation shows highly right-skewed distributions for various healthcare indicators. A few countries have extremely high values compared to the majority, creating long tails in the histograms. This skewness can distort statistical analysis and negatively impact machine learning models.

Log transformation was applied here only to positive values to prevent errors



- After applying the log transformation, the values are now more evenly spread, reducing skewness and making patterns more interpretable. This adjustment helps stabilize variance, making comparisons across countries and improving the performance of our models.

## Handling missing values

To address missing values and maintain data integrity, we applied different strategies:

- We used IQR to detect outliers, so that extreme values were properly flagged without removing any meaningful data points. We capped extreme values instead of removing them to maintain dataset completeness and also to prevent distortion from outliers.
- For lag features, missing values were filled using group-wise mean imputation, replacing NaNs with the average for each country-series to prevent data loss and also preserving historical trends. This ensures realistic estimates without introducing bias.
- For rolling averages, we applied forward fill to replace missing values with the last valid data point to ensure smooth trends and prevent sudden gaps that could distort time-series patterns.
- For year over year growth rate, missing values were filled with 0, assuming no change for those periods. This maintains consistency in trend analysis and prevents any fluctuations in the data.

## Splitting the dataset

The dataset was split into training, validation, and test sets for modeling. However, since the original dataset was reshaped before the split, we reshaped the training set again to ensure consistency in processing. We made sure that all time-series transformations, including lags and rolling averages were applied correctly. Additionally, we computed transformations only on the training set and ensured that missing values were imputed without using validation or test data to prevent data leakage and ensure that no information from these datasets influenced model training.